

Planar

(pre 2011)

FinFET

(2011–2023)

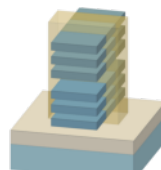
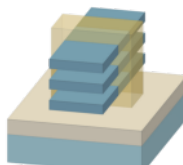
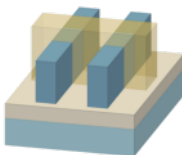
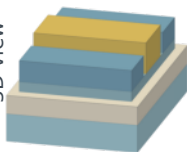
GAAFET

(2023–)

CFET

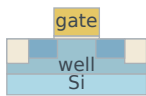
(2032?)

3D view



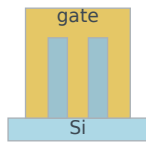
gates ghosted to show
the stacked channels

cross-
section



gate controls one face

$1 \times$ footprint width



gate wraps 3 sides

$\frac{1}{2} \times$ footprint width



gate wraps 4 sides

$\frac{1}{4} \times$ footprint width



two devices, one footprint

$\frac{1}{4} \times$ footprint width

solid backend (semi_structures.solid): true geometry, semi-transparent TiN gates; Si blues, PMOS S/D SiGe, NMOS S/D Si:P, SiO₂ sand